Baseline characteristics of a large multi-site chronic HBV electronic health record-based cohort in the UK

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Introduction

Nearly 300 million people worldwide are chronically infected with hepatitis B virus (HBV), which is the leading cause of global liver cancer deaths (1). The UK HBV population is not well characterised but understanding the burden of disease is crucial to optimise case finding and clinical service provision.

Aim

We aimed to characterise the English CHB population using a representative sample of patients from the QResearch primary care electronic health record (EHR) database in England (2).

Method

We included CHB patients from QResearch, a primary-care database in England of individual-level patient records from 1727 general practices, aged ≥ 18 years with a diagnostic record of CHB between 1999 and 2019. Baseline was defined as the first record of CHB.

Characteristics	n (%), or mean (SD)
Age, years	
18-24	647 (8.1)
25-34	2810 (35.3)
35-44	2443 (30.7)
45-54	1277 (16.1)
55-64	534 (6.7)
65-74	241 (3.0)
Townsend deprivation quintile	
1st (least deprived)	459 (5.8)
2nd	722 (9.1)
3rd	1211 (15.3)
4th	2021 (25.5)
5th (most deprived)	3516 (44.3)
Missing	23 (0.3)
Ethnicity	
Black African	1803 (22.7)
Chinese	865 (10.9)
South Asian	1307 (16.4)
Other	800 (10.1)
White	1894 (23.8)
Alcohol, units/day	
Non-drinker	2564 (32.2)
<1	408 (5.1)
1-2	186 (2.3)
3-6	80 (1.0)
≥7-9	25 (0.4)
Missing	4687 (58.9)
Smoking, cigarettes/day	
Non-smoker	2902 (36.5)
Ex-smoker	662 (8.3)
<10	8648(10.9)
10-19	179 (2.3)
≥20	84(1.1)
Missing	3257 (41.0)
BMI, kg/m^2	
<18.5	98 (1.2)
18.5-24.9	1263 (15.9)
25-29.9	1248 (15.7)
≥30	742 (9.3)
Missing	4601 (57.9)

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Results

We identified 7952 CHB patients (61% male). Mean follow-up was 7.3 years. At baseline, 74% of patients were aged < 45 years, 44% were residing in the most deprived geographical regions, and > 50% were non-white. 1.8%, 2.9% and 3.7% of patients had record of non-alcoholic fatty liver disease (NAFLD), cirrhosis and varices/ascites, respectively. Prevalence of extra-hepatic comorbidities was low, however 9.5% and 5.8% of patients had hypertension and diabetes, respectively. Male sex and ethnicity were significantly associated with deprivation ($p < 0.001$). Within 3 years of diagnosis 145 (1.8%) patients were on antiviral treatment.

Conclusions

The UK CHB morbidity burden is concentrated in young, deprived, non-white individuals among whom a minority are treated. Further research is required to identify factors associated with different disease outcomes and to assess benefits of wider treatment.

References

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